



AI Playbook and Toolkit for Public Health Departments

AI Investment Strategy and Business Case Review

EXE 420

AI investments compete with many other public health priorities. Leaders need a disciplined way to decide when AI is worth funding, what costs must be considered, and how benefits will be measured. A business case review helps agencies avoid buying tools that are impressive in demonstrations but weak in operational value, sustainability, interoperability, or public health impact. This module teaches leaders to evaluate AI investments across cost, value, readiness, risk, staffing, procurement, lifecycle support, and sustainability. The business case should not focus only on software price. It should include data preparation, implementation, integration, training, governance, validation, monitoring, maintenance, vendor management, cybersecurity, accessibility, and eventual replacement or retirement.

Level	advanced/leadership
Primary track	Public Health Executive Leadership Track
Appears in tracks	governance-security

Learning Objectives

- Identify the full cost components of AI implementation and operations.
- Evaluate expected benefits in terms of public health value, workflow improvement, equity, quality, timeliness, and staff burden.
- Distinguish between a vendor demonstration and a credible business case.
- Develop business case questions for AI procurement or internal development.
- Produce an AI business case review worksheet.

Module Overview

AI investments compete with many other public health priorities. Leaders need a disciplined way to decide when AI is worth funding, what costs must be considered, and how benefits will be measured. A business case review helps agencies avoid buying tools that are impressive in demonstrations but weak in operational value, sustainability, interoperability, or public health impact.

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Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Public health agencies often face short-term funding cycles and urgent operational needs. AI products may be marketed as quick solutions, but the true cost of responsible implementation is broader than subscription fees. Agencies may need to update data pipelines, configure security controls, develop validation plans, train staff, write policies, monitor performance, and manage vendor relationships.

A strong business case also protects mission focus. AI should not be funded because it is novel. It should be funded when it addresses a meaningful public health problem, fits agency capacity, can be governed responsibly, and can produce measurable value. Business case review helps leaders compare AI investments with non-AI alternatives, such as workflow redesign, staffing changes, data quality improvements, or simpler automation.

Public Health Example

A vendor offers an AI tool that promises to reduce disease investigator workload by summarizing case reports. The annual license appears affordable, but the agency must also pay for integration with the surveillance system, security review, staff training, validation, legal review, vendor support, monitoring dashboards, and staff time for governance. If those costs are not considered, leadership may underestimate the investment required.

A business case review asks whether the tool will meaningfully reduce burden, improve timeliness, maintain quality, and fit the agency's workflow. It also asks whether a simpler approach, such as better case report templates or workflow redesign, would solve the problem at lower cost and risk. The review does not assume AI is the answer; it tests whether AI is the right answer.

Technical and Governance Deep Dive

An AI business case should begin with a problem statement. The problem should be operational and measurable, such as delayed case review, inconsistent public response drafting, high manual data cleanup burden, or inability to search internal guidance efficiently. A vague desire to 'use AI' is not a sufficient problem statement. The business case should connect the proposed AI use to a real public health need.

The review should compare alternatives. Options may include doing nothing, improving workflow, improving data quality, using rules-based automation, procuring a vendor product, developing an internal tool, or joining a shared service. Each option should be assessed for benefit, cost, risk, time, staffing, sustainability, and governance burden. This comparison prevents AI from becoming the default answer to every operational problem.

Success measures should be defined before funding. Measures might include reduced processing time, improved completeness, fewer manual touches, better routing accuracy, staff satisfaction, fewer errors, improved language access, or reduced backlog. Measures should also include safety and equity indicators. A tool that saves time but increases error rates, widens disparities, or erodes public trust should not be considered successful.

Risks, Failure Modes, and Guardrails

One risk is hidden cost. Agencies may purchase an AI tool without funding integration, monitoring, validation, staff training, or ongoing support. A guardrail is to require total cost of ownership in every business case.

Another risk is benefits inflation. Vendors or project champions may overstate time savings or accuracy without local evidence. A guardrail is to require pilot success criteria and benefits realization review before scaling.

Application to AI-Supported Workflows

Generative AI business cases should include review time, source verification, and records considerations. Predictive analytics business cases should include validation data, subgroup testing, and monitoring costs. Agentic AI business cases should include security architecture, approval workflows, rollback, logging, and incident response. Cost and value must reflect responsible use.

Reflection Questions

Which current or proposed AI use case in your agency raises this leadership or governance issue most clearly?

Who currently has authority to make decisions about this issue, and is that authority documented?

What evidence would governance or leadership need before approving the use case?

Which staff, partners, or communities would be affected if this issue were handled poorly?

What policy, process, or artifact should be created or updated after this module?

Practical Exercise

Complete an AI business case review worksheet for one proposed investment. Include problem, alternatives, expected benefits, total cost of ownership, risk, equity considerations, implementation needs, and success measures.

Before You Begin

Choose a real or realistic public health AI use case. Document assumptions, unresolved questions, and which agency roles would need to review the artifact before it is adopted. The exercise should produce a work product that could support governance review, leadership discussion, implementation planning, or policy development.

Expected Artifact or Evidence

AI business case worksheet; total cost of ownership estimate; benefits realization plan; alternatives analysis.

Expected Assignment

- AI business case worksheet; total cost of ownership estimate; benefits realization plan; alternatives analysis.

Knowledge Check

- 1. What is the strongest reason this module topic belongs in AI leadership and governance training?
- 2. Which practice best supports accountable public health AI governance?
- 3. When should an AI use case be escalated for leadership or governance review?
- 4. What is weak evidence of completion for a leadership and governance module?
- 5. What should leaders do when an AI use case has meaningful benefits but unresolved risks?

References and Resources

- NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0):
<https://www.nist.gov/itl/ai-risk-management-framework>
- NIST Generative AI Profile: <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>
- NIST Privacy Framework
- CDC Public Health Data Strategy and Data Modernization resources:
<https://www.cdc.gov/public-health-data-strategy/php/about/index.html>
- WHO Ethics and Governance of Artificial Intelligence for Health:
<https://www.who.int/publications/i/item/9789240029200>
- OMB Memorandum M-24-10 on federal AI governance, risk management, and innovation, where relevant to public-sector governance models
- Agency strategic plans, procurement policies, privacy policies, cybersecurity policies, records policies, and equity plans